



FACULTY OF SCIENCE AND HUMANITIES
SRM INSTITUTE OF SCIENCE AND TECHNOLOGY



7. Programme Structure						
Discipline Core Courses (C)				Discipline Core/Elective Courses (D)		
Course Code	Course Title	Session / Week			C	
		L	T	P		
UDS24101J	Programming using Java	3	0	3	4	
UDS24102J	Fundamentals of Data Science	3	0	3	4	
UMS24103T	Mathematics for Artificial Intelligence	4	0	0	4	
UDS24201J	Elements of Distributed Data Processing	3	0	3	4	
UDS24202J	Data Structures and Algorithms	3	0	3	4	
UMS24203T	Statistics for Artificial Intelligence	4	0	0	4	
UDS24301J	Data Engineering for Enterprises	3	0	3	4	
UDS24302J	Database Management System	3	0	3	4	
UDS24303J	Natural Language Processing	3	0	3	4	
UDS24401J	Deep Learning	3	0	3	4	
Total Learning Credits					40	
UDS24D01J	Advanced Computing with Python	3	0	2	4	
UDS24D02J	Machine Learning					
UDS24D03J	Deep Learning with Keras and Tensorflow	3	0	3	4	
UDS24D04J	Big Data Analytics with Applications					
UDS24D05J	Intelligent Automation	3	0	3	4	
UDS24D06J	Computer Vision					
UDS24D07J	Advanced Analytics and Data Visualization	3	0	3	4	
UDS24D08J	Data Science for Business Analytics					
UDS24D09J	Artificial Intelligence and Automation for Enterprises	3	0	3	4	
UDS24D10J	Data Wrangling					
UDS24D11T	Research Methodology in Data Science	4	0	0	4	
UDS24D12T	Data Analytics for Project Management					
UDS24D13J	Data Warehousing and Data Mining	3	0	3	4	
UDS24D14J	Cloud and Grid Computing					
UDS24D15J	Machine Learning for Enterprises	3	0	3	4	
UDS24D16J	Real World Computer Vision Applications					
UDS24D17J	Technology Leadership and Innovation Management	3	0	2	4	
UDS24D18J	Social Media and Text Analytics					
UDS24D19T	Statistical Analysis and Business Applications	4	0	0	4	
UDS24D20T	Applications of Edge IoT and Machine Learning					
Total Learning Credits					40	

Elective Courses (Minor) (E)						
Course Code	Course Title	Session/Week			C	
		L	T	P		
UDS24E01J	Data Science and Analytics	3	0	3	4	
UDS24E02J	Data Analytics using Spreadsheet	3	0	3	4	
UDS24E03J	Essentials of Machine Learning	3	0	3	4	
UDS24E04J	Programming using R	3	0	3	4	
UDS24E05J	Programming using Python	3	0	3	4	
UDS24E06J	Introduction to Deep Learning	3	0	3	4	
UDS24E07J	Applications of Computer Vision	3	0	3	4	
UDS24E08J	Data Visualization Tools	3	0	3	4	
Total Learning Credits					32	

Multidisciplinary Courses (M)						
Course Code	Course Title	Session / Week			C	
		L	T	P		
UDS24M01J	Internet of Things	2	0	2	3	
UDS24M02J	Digital Transformation	2	0	2	3	
UDS24M03J	Blockchain Technology	2	0	3	3	
Total Learning Credits					9	

Value Added Courses (V)						
Course Code	Course Title	Hours/Week			C	
		L	T	P		
UCD24V01T	Essentials of Artificial Intelligence	1	0	0	1	
UES24V01T	Environmental Studies	2	0	0	2	
UCD24V02T	Universal Human Values	2	0	0	2	
Total Learning Credits					5	

Ability Enhancement Courses (A)						
Course Code	Course Title	Session/Week			C	
		L	T	P		
ULT24AE1J	Tamil – I	2	0	2	3	
ULH24AE1J	Hindi - I					
ULF24AE1J	French - I					
ULT24AE2J	Tamil – II	2	0	2	3	
ULH24AE2J	Hindi - II					
ULF24AE2J	French - II					
ULE24AE2J	Business English	2	0	2	3	
Total Learning Credits					9	

Skill Enhancement Courses (S)						
Course Code	Course Title	Session/Week			C	
		L	T	P		
UCD24S01J	Verbal Ability and Skill Development	1	0	2	2	
UCD24S02L	Quantitative Aptitude and Logical Reasoning	0	0	2	1	
UCD24S03J	Industry Oriented Employability and Leadership Skills	1	0	2	2	
UEN24S01L	Communication Skills	0	0	4	2	
UCD24S04J	Career Readiness and Professional Skills	1	0	2	2	
Total Learning Credits					9	

ULF24AE2J	French - II				
ULE24AE2J	Business English	2	0	2	3
Total Learning Credits					9

	Leadership Skills				
UEN24S01L	Communication Skills	0	0	4	2
UCD24S04J	Career Readiness and Professional Skills	1	0	2	2
Total Learning Credits					9

6/200



Bachelor's Programme, Regulations 2024

5



FACULTY OF SCIENCE AND HUMANITIES
SRM INSTITUTE OF SCIENCE AND TECHNOLOGY



Project Work / Internship (P)					
Course Code	Course Title	Session/Week			C
		L	T	P	
UDS24P01L	Internship – I	0	0	0	1
UDS24P02L	Internship – II	0	0	0	1
UDS24P03L	Project Work	0	0	4	2
UDS24P04L	Research Project and Dissertation	0	0	24	12
UDS24P05L	Professional Internship				
Total Learning Credits					16

Mandatory Courses (Y)					
Course Code	Course Title	Session/Week			C
		L	T	P	
UNS24Y01L	NSS				
UNC24Y01L	NCC				
UNO24Y01L	NSO	0	0	0	0
UYG24Y01L	YOGA				
UMI24Y01L	My India Project	0	0	0	0
Total Learning Credits					0

Internship - I to be completed during summer vacation of the first year

Internship - II to be completed during summer vacation of the second year

Students from other departments can enroll in Elective Courses (Minor) (E)

Students from all departments will be able to enroll in Multidisciplinary Courses (M).



8. Course Allocation across Semesters

Semester – I					
Course Code	Course Title	Sessions Per Week			C
		L	T	P	
ULT24AE1J	Tamil – I	2	0	2	3
ULH24AE1J	Hindi – I				
ULF24AE1J	French – I				
ULE24AE2J	Business English	2	0	2	3
UDS24101J	Programming using Java	3	0	3	4
UDS24102J	Fundamentals of Data Science	3	0	3	4
UMS24103T	Mathematics for Artificial Intelligence	4	0	0	4
UCD24S01J	Verbal Ability and Skill Development	1	0	2	2
Total Learning Credits					20

Semester – II					
Course Code	Course Title	Sessions Per Week			C
		L	T	P	
ULT24AE2J	Tamil – II	2	0	2	3
ULH24AE2J	Hindi – II				
ULF24AE2J	French – II				
UDS24201J	Elements of Distributed Data Processing	3	0	3	4
UDS24202J	Data Structures and Algorithms	3	0	3	4
UMS24203T	Statistics for Artificial Intelligence	4	0	0	4
UCD24V01T	Essentials of Artificial Intelligence	1	0	0	1
	Multidisciplinary Course – I				3
UCD24S02L	Quantitative Aptitude and Logical Reasoning	0	0	2	1
UNS24Y01L	NSS	0	0	0	0
UNC24Y01L	NCC				
UNO24Y01L	NSO				
UYG24Y01L	YOGA				
Total Learning Credits					20

Total Learning Credits of Courses of FIRST year = 40

8/200



FACULTY OF SCIENCE AND HUMANITIES
SRM INSTITUTE OF SCIENCE AND TECHNOLOGY



Semester – III					
Course Code	Course Title	Sessions Per Week			C
		L	T	P	
UDS24301J	Data Engineering for Enterprises	3	0	3	4
UDS24302J	Database Management System	3	0	3	4
UDS24303J	Natural Language Processing	3	0	3	4
	Minor Elective – I				4
UDS24P01L	Internship - I	0	0	0	1
	Multidisciplinary Course – II				3
Total Learning Credits					20

Semester – IV					
Course Code	Course Title	Sessions Per Week			C
		L	T	P	
UDS24401J	Deep Learning	3	0	3	4
UDS24D01J	Advanced Computing with Python	3	0	2	4
UDS24D02J	Machine Learning				
	Minor Elective – II				4
	Minor Elective – III				4
UEN24S01L	Communication Skills	0	0	4	2
UCD24S03J	Industry Oriented Employability and Leadership Skills	1	0	2	2
UMI24Y01L	My India Project	0	0	0	0
Total Learning Credits					20

Total Learning Credits of Courses of SECOND year = 40

Semester – V					
Course Code	Course Title	Sessions Per Week			C
		L	T	P	
UDS24D03J	Deep Learning with Keras and Tensorflow	3	0	3	4
UDS24D04J	Big Data Analytics with Applications				
UDS24D05J	Intelligent Automation				
UDS24D06J	Computer Vision	3	0	3	4
	Minor Elective – IV				
	Multidisciplinary Course – III				
UES24V01T	Environmental Studies	2	0	0	2
UCD24S04J	Career Readiness and Professional Skills	1	0	2	2
UDS24P02L	Internship - II	0	0	0	-
Total Learning Credits					

10/200



Semester – VI					
Course Code	Course Title	Sessions Per Week			C
		L	T	P	
UDS24D07J	Advanced Analytics and Data Visualization	3	0	3	4
UDS24D08J	Data Science for Business Analytics				
UDS24D09J	Artificial Intelligence and Automation for Enterprises	3	0	3	4
UDS24D10J	Data Wrangling				
	Minor Elective – V				4
	Minor Elective – VI				4
UDS24P03L	Project Work	0	0	4	2
UCD24V02T	Universal Human Values	2	0	0	2
Total Learning Credits					20

Total Learning Credits of Courses of THIRD year = 40



Semester – VII					
Course Code	Course Title	Sessions Per Week			C
		L	T	P	
UDS24D11T	Research Methodology in Data Science	4	0	0	4
UDS24D12T	Data Analytics for Project Management				
UDS24D13J	Data Warehousing and Data Mining	3	0	3	4
UDS24D14J	Cloud and Grid Computing				
UDS24D15J	Machine Learning for Enterprises	3	0	3	4
UDS24D16J	Real World Computer Vision Applications				
	Minor Elective – VII				4
	Minor Elective – VIII				4
Total Learning Credits					20

Semester – VIII					
Course Code	Course Title	Sessions Per Week			C
		L	T	P	
UDS24D17J	Technology Leadership and Innovation Management	3	0	2	4
UDS24D18J	Social Media and Text Analytics				
UDS24D19T	Statistical Analysis and Business Applications	4	0	0	4
UDS24D20T	Applications of Edge IoT and Machine Learning				
UDS24P04L	Research Project and Dissertation	0	0	24	12
UDS24P05L	Professional Internship				
Total Learning Credits					20

Total Learning Credits of Courses of FOURTH year = 40



FACULTY OF SCIENCE AND HUMANITIES
SRM INSTITUTE OF SCIENCE AND TECHNOLOGY



8.1 Elective Courses (MINOR) offered by the Department of Computer Applications to other Departments (excluding Department of Computer Applications)

Elective Courses (Minor) (E)						
Semester	Course Code	Course Title	Sessions Per Week			C
			L	T	P	
III	UDS24E01J	Data Science and Analytics	3	0	3	4
IV	UDS24E02J	Data Analytics using Spreadsheet	3	0	3	4
IV	UDS24E03J	Essentials of Machine Learning	3	0	3	4
V	UDS24E04J	Programming using R	3	0	3	4
VI	UDS24E05J	Programming using Python	3	0	3	4
VI	UDS24E06J	Introduction to Deep Learning	3	0	3	4
VII	UDS24E07J	Applications of Computer Vision	3	0	3	4
VII	UDS24E08J	Data Visualization Tools	3	0	3	4
Total Learning Credits						32

8.2 Multidisciplinary Courses offered by the Department of Computer Applications to all Departments (including Department of Computer Applications)

Multidisciplinary Courses (M)						
Semester	Course Code	Course Title	Sessions Per Week			C
			L	T	P	
II	UDS24M01J	Internet of Things	2	0	2	3
III	UDS24M02J	Digital Transformation	2	0	2	3
V	UDS24M03J	Blockchain Technology	2	0	3	3
Total Learning Credits						9